STPM Physics · Semester 3 · Chapter 20

OSCILLATIONS

20.1 Characteristics of Simple Harmonic Motion

DEFINITIONS & LAWS

- **Simple harmonic motion** is the motion of a particle whose acceleration is directly proportional to its displacement from a fixed (equilibrium) point and is always directed towards that point.
- **Amplitude** is the maximum displacement of the particle from the equilibrium position (m).
- **Equilibrium position** is the point at which no resultant force acts on the body performing SHM; displacement is measured from this point.
- **Period** is the time taken for one complete oscillation; **frequency** is the number of complete oscillations per unit time.
- **Angular frequency** is the rate of change of phase angle, $\omega = 2\pi f = \frac{2\pi}{T}$ (rad s^{-1}).
- **Phase difference** is the fraction of a cycle by which one oscillation is ahead of or behind another, expressed in radian or degree.
- In SHM the restoring force is directly proportional to the displacement and opposite in direction to it. Velocity leads displacement by $\frac{\pi}{2}$ rad; acceleration is in antiphase (π rad) with displacement.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Defining equation of SHM	$a = -\omega^2 x$	a = acceleration (m s^{-2}); ω = angular frequency (rad s^{-1}); x = displacement (m)
Restoring force	$F = -kx = -m\omega^2 x$	k = force constant (N m^{-1}); m = mass (kg); x = displacement (m)
Angular frequency	$\omega = 2\pi f = \frac{2\pi}{T}$	f = frequency (Hz); T = period (s)
Maximum velocity	$v_{max} = \omega A$	At x = 0; A = amplitude (m)
Maximum acceleration	$a_{max} = \omega^2 A$	At x = ±A; A = amplitude (m)

- For a graph of a against x: a straight line through the origin with gradient $-\omega^2$.

TIPS TO ANSWER

- Quote both conditions in a definition: acceleration ∝ displacement AND directed towards the equilibrium position (minus sign).
- Describe a graph of a against x as a straight line through the origin with negative gradient; the magnitude of the gradient is ω^2.
- State the position of each extreme: at x = ±A, v = 0 and |a| and |F| are maximum; at x = 0, |v| is maximum and a = F = 0.
- For a phase relationship, give the angle with the direction: velocity leads displacement by $\frac{\pi}{2}$ rad.
- To show that a motion is SHM from an equation such as x = A sin ωt (or A cos ωt, or one with a phase angle), differentiate twice to obtain v and then a, and show that $a = -\omega^2 x$.

SECTION A TRAPS

Trap: Speed is maximum at the extreme positions. Wrong: the particle momentarily stops at the extremes; speed is maximum at the equilibrium position.

Trap: Reading the magnitude of the gradient of the a–x graph as ω or A. The gradient is $-\omega^2$, so $\omega = \sqrt{|\text{gradient}|}$.

Trap: Taking acceleration to be constant or the restoring force independent of displacement; both contradict $a = -\omega^2 x$.

TRICKS

Trick: Velocity and acceleration are a quarter cycle apart in their extremes: when one is maximum, the other is zero.

Trick: Quick check on any SHM statement: if it says 'constant' or 'independent of displacement' for a or F, it is not SHM.

20.2 Kinematics of Simple Harmonic Motion

DEFINITIONS & LAWS

- **Displacement** in SHM is a sinusoidal function of time; for a particle starting from the equilibrium position, x = A sin ωt.
- Velocity is the rate of change of displacement, $v = \frac{dx}{dt}$, and acceleration is the rate of change of velocity, $a = \frac{dv}{dt} = -\omega^2 x$.
- The speed at displacement x is given by $v = \pm\omega\sqrt{A^2 - x^2}$; the speed is zero at x = ±A and maximum at x = 0.

- The period of SHM is independent of the amplitude.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Displacement	$x = A \sin \omega t$; $x = A \cos \omega t$	A = amplitude (m); sine form starts at equilibrium; cosine form starts at the extreme position
Velocity	$v = \frac{dx}{dt} = \omega A \cos \omega t$	v = velocity (m s^{-1})
Acceleration	$a = -\omega^2 A \sin \omega t = -\omega^2 x$	a = acceleration (m s^{-2})
Velocity at displacement x	$v = \pm\omega\sqrt{A^2 - x^2}$	x = displacement (m); A = amplitude (m)
Maximum values and period	$v_{max} = \omega A$; $a_{max} = \omega^2 A$; $T = \frac{2\pi}{\omega}$	v_{max} = maximum velocity (m s^{-1}); a_{max} = maximum acceleration (m s^{-2}); T = period (s)

- Calculator must be in radian mode when evaluating sin ωt and cos ωt.

TIPS TO ANSWER

- Identify the starting condition first: x = 0 at t = 0 gives sine; x = A at t = 0 gives cosine.
- State formula, substitute with units, give the answer to 3 s.f. with unit (and direction when velocity is required).
- To find the time to reach a displacement, solve $\sin \omega t = \frac{x}{A}$ for ωt in radian, then $t = \frac{\omega t}{\omega}$.
- When the period is given, convert first: $\omega = \frac{2\pi}{T}$. When the amplitude is in cm, convert to m before using $v_{max} = \omega A$.

SECTION A TRAPS

Trap: Using degree mode for ωt (ωt is in radian). The wrong mode gives a displacement and velocity far from the true values.

Trap: Using $v_{max} = \omega^2 A$ or $a_{max} = \omega A$. Check the units: ωA is in m s^{-1}, $\omega^2 A$ is in m s^{-2}.

Trap: Forgetting that $\omega = \frac{2\pi}{T}$, so quoting $v_{max} = \frac{A}{T}$ or using the frequency f in place of ω.

Trap: Using $v = \omega(A^2 - x^2)$ without the square root.

TRICKS

Trick: Time to reach $\frac{A}{2}$ from equilibrium is $\frac{T}{12}$ (phase $\frac{\pi}{6}$); time to reach A is $\frac{T}{4}$.

Trick: Check v at x: $\frac{v}{v_{max}} = \sqrt{1 - (\frac{x}{A})^2}$. For x = 0.6A, v = 0.8$v_{max}$ (3-4-5 triangle).

20.3 Energy in Simple Harmonic Motion

DEFINITIONS & LAWS

- **Total energy** of a particle in undamped SHM is constant and is equal to its maximum kinetic energy (at the equilibrium position) or its maximum potential energy (at the extreme positions).
- Kinetic energy is maximum at x = 0 and zero at x = ±A; potential energy is zero at x = 0 and maximum at x = ±A.
- Energy is continuously interchanged between kinetic and potential forms with the total remaining constant when there is no damping.
- The total energy is directly proportional to the square of the amplitude.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Total energy	$E = \frac{1}{2}m\omega^2A^2 = \frac{1}{2}kA^2$	m = mass (kg); ω = angular frequency (rad s^{-1}); A = amplitude (m); k = $m\omega^2$ (N m^{-1})
Kinetic energy	$E_k = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$	E_k = kinetic energy (J); v = velocity (m s^{-1}); x = displacement (m)
Potential energy	$E_p = \frac{1}{2}m\omega^2x^2 = \frac{1}{2}kx^2$	E_p = potential energy (J); x = displacement (m)
Conservation of energy	$E_k + E_p = E$	E is constant (J)
Fractions of total energy	$\frac{E_k}{E} = 1 - (\frac{x}{A})^2$; $\frac{E_p}{E} = (\frac{x}{A})^2$	Ratios of kinetic and potential energy to the total energy; x = displacement (m); A = amplitude (m)

TIPS TO ANSWER

- Sketch of E against x: E_p is a parabola with minimum at x = 0; E_k is an inverted parabola; E is a horizontal line at $\frac{1}{2}kA^2$; the curves cross where $E_k = E_p$.
- Sketch of energy against time: E_k and E_p have frequency 2f (period $\frac{T}{2}$) and are in antiphase.

- State: $E = \frac{1}{2}kA^2$ → substitute with amplitude in m → answer in joule.
- For fractions, write $\frac{E_p}{E} = (\frac{x}{A})^2$ and substitute the ratio directly; no need for m or ω.

SECTION A TRAPS

Trap: Taking energy to be proportional to A. The total energy is proportional to A^2: doubling the amplitude gives four times the energy.

Trap: Equal kinetic and potential energies at $x = \frac{A}{2}$. Wrong: $E_k = E_p$ when $x^2 = \frac{A^2}{2}$, i.e. $x = \frac{A}{\sqrt{2}} \approx 0.707A$.

Trap: Omitting the factor ½ in $E = \frac{1}{2}kA^2$, or using the displacement x instead of the amplitude A for the total energy.

Trap: Assuming $E_k = E_p$ at $x = \frac{A}{2}$ gives 50%: at $x = \frac{A}{2}$, E_p = 25% and E_k = 75% of the total.

TRICKS

Trick: Use fractions: $x = \frac{A}{2}$ gives $E_p = \frac{1}{4}E$; $x = \frac{A}{4}$ gives $E_p = \frac{1}{16}E$; $x = \frac{A}{\sqrt{2}}$ gives ½E.

Trick: Scaling: $E \propto A^2$ (same k), so tripling A gives 9 times the energy.

20.4 Systems in Simple Harmonic Motion

DEFINITIONS & LAWS

- A **mass–spring system** performs SHM because the restoring force on the mass is directly proportional to its extension and is directed towards the equilibrium position (Hooke's law, $F = -kx$).
- A **simple pendulum** performs SHM only for small angular displacement ($\sin\theta \approx \theta$ in radian, up to about 10°); the restoring force along the arc is $mg\sin\theta \approx mg\theta$.
- The period of a simple pendulum is independent of the mass of the bob and of the amplitude (for small angles).
- The period of a mass–spring system is independent of the amplitude and, for a vertical spring, of g.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Mass–spring period	$T = 2\pi\sqrt{\frac{m}{k}}$	T = period (s); m = mass (kg); k = force constant (N m^{-1})
Mass–spring angular frequency	$\omega = \sqrt{\frac{k}{m}}$	ω = angular frequency (rad s^{-1}); k = force constant (N m^{-1}); m = mass (kg)
Simple pendulum period	$T = 2\pi\sqrt{\frac{L}{g}}$	L = length from the point of suspension to the centre of the bob (m); g = acceleration of free fall (m s^{-2})
Springs in parallel	$k = k_1 + k_2$	k = effective force constant (N m^{-1}); k_1, k_2 = force constants of the springs (N m^{-1})
Springs in series	$\frac{1}{k} = \frac{1}{k_1} + \frac{1}{k_2}$	k = effective force constant (N m^{-1}); k_1, k_2 = force constants of the springs (N m^{-1})
Proportionalities for period	$T \propto \sqrt{m}$; $T \propto \sqrt{L}$; $T \propto \frac{1}{\sqrt{g}}$; $T \propto \frac{1}{\sqrt{k}}$	T = period (s); m = mass (kg); L = length (m); g = acceleration of free fall (m s^{-2}); k = force constant (N m^{-1})

- g = 9.81 m s^{-2} on Earth.

TIPS TO ANSWER

- State the condition for SHM: small angle (pendulum), or Hooke's law obeyed (spring).
- Use the ratio method for changes: $\frac{T_2}{T_1} = \sqrt{\frac{m_2}{m_1}}$ or $\sqrt{\frac{L_2}{L_1}}$ or $\sqrt{\frac{g_1}{g_2}}$; show the ratio before the final answer.
- Derive the period by showing $a = -\frac{k}{m}x$, then compare with $a = -\omega^2 x$ to give $\omega^2 = \frac{k}{m}$.
- Give the period in seconds to 3 s.f.; if frequency is asked, take $f = \frac{1}{T}$.

SECTION A TRAPS

Trap: Writing $T = 2\pi\sqrt{\frac{g}{L}}$ or $T = 2\pi\sqrt{\frac{k}{m}}$: inverted ratios. Check the unit: $\sqrt{\frac{m}{k}}$ and $\sqrt{\frac{L}{g}}$ both give seconds.

Trap: Believing a heavier bob lengthens the pendulum period. The pendulum period is independent of the mass.

Trap: Using the mass-spring result with the wrong sense: quadrupling the mass doubles the period ($T \propto \sqrt{m}$), it does not quadruple it.

Trap: On the Moon (smaller g) the pendulum period is longer, not shorter, since $T \propto \frac{1}{\sqrt{g}}$.

TRICKS

Trick: Quadruple the length or mass: period doubles. A period ratio of 2 means a length ratio of 4.

Trick: Springs in parallel make the system stiffer, so the period decreases; in series the period increases.

20.5 Damped Oscillation

DEFINITIONS & LAWS

- **Damped oscillation** is an oscillation in which the amplitude decreases with time because energy is dissipated (as heat) by resistive forces such as air resistance or friction.
- **Light (under) damping**: the system oscillates with amplitude decreasing exponentially with time; the period is slightly longer than that of the free oscillation (almost unchanged when the damping is light).
- **Critical damping**: the system returns to the equilibrium position in the shortest possible time without oscillating.
- **Heavy (over) damping**: the system returns to the equilibrium position slowly without oscillating.
- Explain damping: the resistive force does negative work on the oscillator, so the total mechanical energy decreases and the amplitude decreases.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Amplitude decay	$A = A_0 e^{-bt}$	A_0 = initial amplitude (m); b = damping constant (s^{-1}); t = time (s)
Energy decay	$E \propto A^2$; $E = E_0 e^{-2bt}$	E = energy (J); E_0 = initial energy (J); b = damping constant (s^{-1}); t = time (s)
Exponential decay of amplitude	$\frac{A_2}{A_1} = \frac{A_1}{A_0}$	For equal time intervals the amplitude falls by the same factor (exponential decay); A_0, A_1, A_2 = amplitudes at successive equal intervals (m)
Half-life of amplitude	$t_{½} = \frac{\ln 2}{b}$	$t_{½}$ = half-life of amplitude (s); b = damping constant (s^{-1})

TIPS TO ANSWER

- For a displacement–time graph of light damping, sketch oscillations whose peaks lie on an exponentially decaying envelope; label both axes.
- For critical and heavy damping, sketch a curve that falls to zero without crossing the time axis; critical damping reaches zero faster than heavy damping.
- Always state the energy change: mechanical energy is converted to thermal energy by the resistive force.
- Give a use of each type: critical damping in a car suspension, a galvanometer and door closers.
- When asked to compare free, light, critical and heavy damping on **one** displacement–time graph, start **every** curve from the amplitude (x = A at t = 0). The comparison refers to an experiment in which the system is pulled to its amplitude and released. Free oscillation: constant amplitude and period; light damping: decaying amplitude; critical damping: returns to zero fastest without crossing; heavy damping: returns slowly without crossing.

SECTION A TRAPS

Trap: Stating that the total energy stays constant while only the amplitude changes. Energy decreases as A^2 decreases.

Trap: Assuming the amplitude falls by a constant amount each cycle. It falls by a constant fraction (exponential).

Trap: Saying that heavy damping returns the system to equilibrium fastest. Critical damping is the fastest without oscillation.

Trap: Saying that damping greatly increases the frequency; for light damping the period is almost unchanged.

TRICKS

Trick: Equal ratios: 0.12 → 0.075 gives a ratio of 0.625, so the next equal interval gives 0.075 × 0.625.

Trick: Halving every t_1 seconds: after n intervals the amplitude is $\frac{A_0}{2^n}$; the energy is $\frac{{A_0}^2}{4^n}$.

20.6 Forced Oscillation and Resonance

DEFINITIONS & LAWS

- **Free oscillation** is the oscillation of a system without any external force acting on it after the initial disturbance; no energy is lost, so the amplitude is constant and the system oscillates at its natural frequency.

- **Forced oscillation** is an oscillation of a system driven by an external periodic force; it oscillates at the frequency of the driving force.
- **Resonance** occurs when the driving frequency is equal to the natural frequency of the system, giving maximum amplitude of oscillation (maximum energy transfer to the system).
- Effect of damping: increasing damping reduces the amplitude at resonance and makes the resonance curve broader (less sharp); the peak shifts slightly to lower frequency.
- At resonance the driving force leads the displacement by $\frac{\pi}{2}$ rad.

FORMULAE

Quantity / law	Formula	Symbols, units & conditions
Natural frequency, mass–spring	$f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$	f_0 = natural frequency (Hz); k = force constant (N m^{-1}); m = mass (kg)
Natural frequency, simple pendulum	$f_0 = \frac{1}{2\pi}\sqrt{\frac{g}{L}}$	f_0 = natural frequency (Hz); g = acceleration of free fall (m s^{-2}); L = length (m)
Resonance condition	$f_{driving} = f_0$	$f_{driving}$ = frequency of the driving force (Hz); f_0 = natural frequency (Hz)

- Graph: amplitude against driving frequency, with a peak at f_0.

TIPS TO ANSWER

- For the resonance graph, label axes (amplitude against driving frequency), draw a peak at f_0 and show a second curve for heavier damping which is lower and broader.
- Explain resonance in two points: driving frequency equals natural frequency; energy transferred to the system at the maximum rate, so the amplitude is maximum.
- Give examples with a consequence: collapse of a bridge or a building in an earthquake; useful: tuning of a radio circuit, microwave oven.
- Calculation: find $f_0 = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$ and state that resonance occurs at this driving frequency.

SECTION A TRAPS

Trap: Leaving out the 2π, i.e. quoting $\sqrt{\frac{k}{m}} = 22$ as a frequency in Hz. This is ω in rad s^{-1}; $f = \frac{\omega}{2\pi}$.

Trap: Believing that more damping gives a higher and sharper resonance peak. More damping gives a lower, broader peak.

Trap: Saying that resonance occurs at the largest driving force or the largest driving frequency; it depends on the match with f_0.

Trap: Confusing resonance with beats or interference.

TRICKS

Trick: If options include $\sqrt{\frac{k}{m}}$ and $\sqrt{\frac{k}{m}}/2\pi$, the frequency in Hz is the one divided by 2π; the other is ω.

Trick: Limiting case: no damping gives an infinite peak; heavy damping gives no distinct peak.